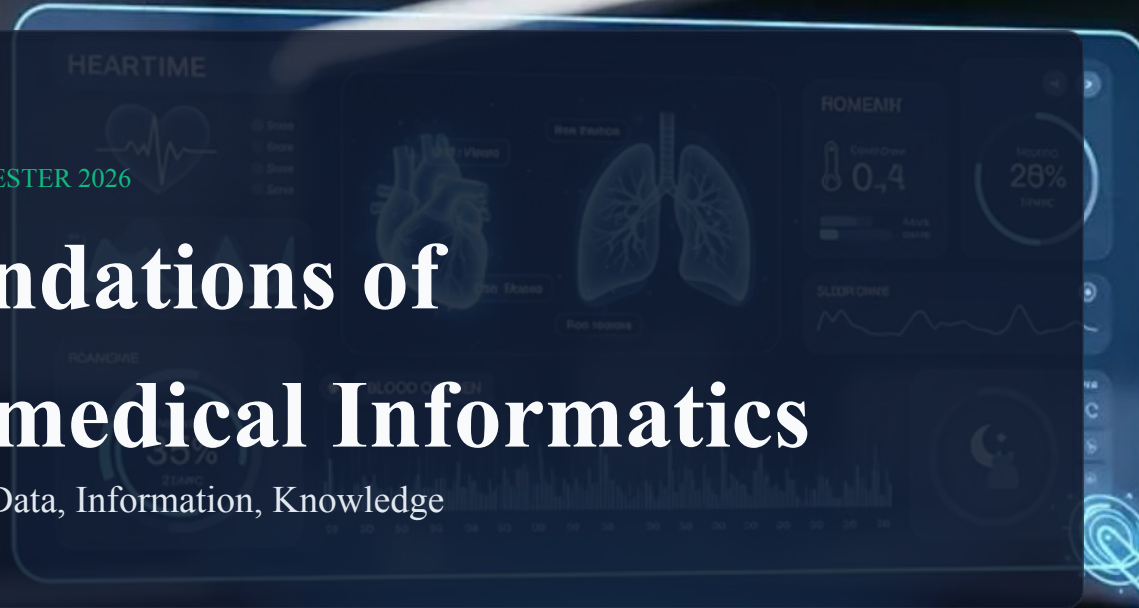


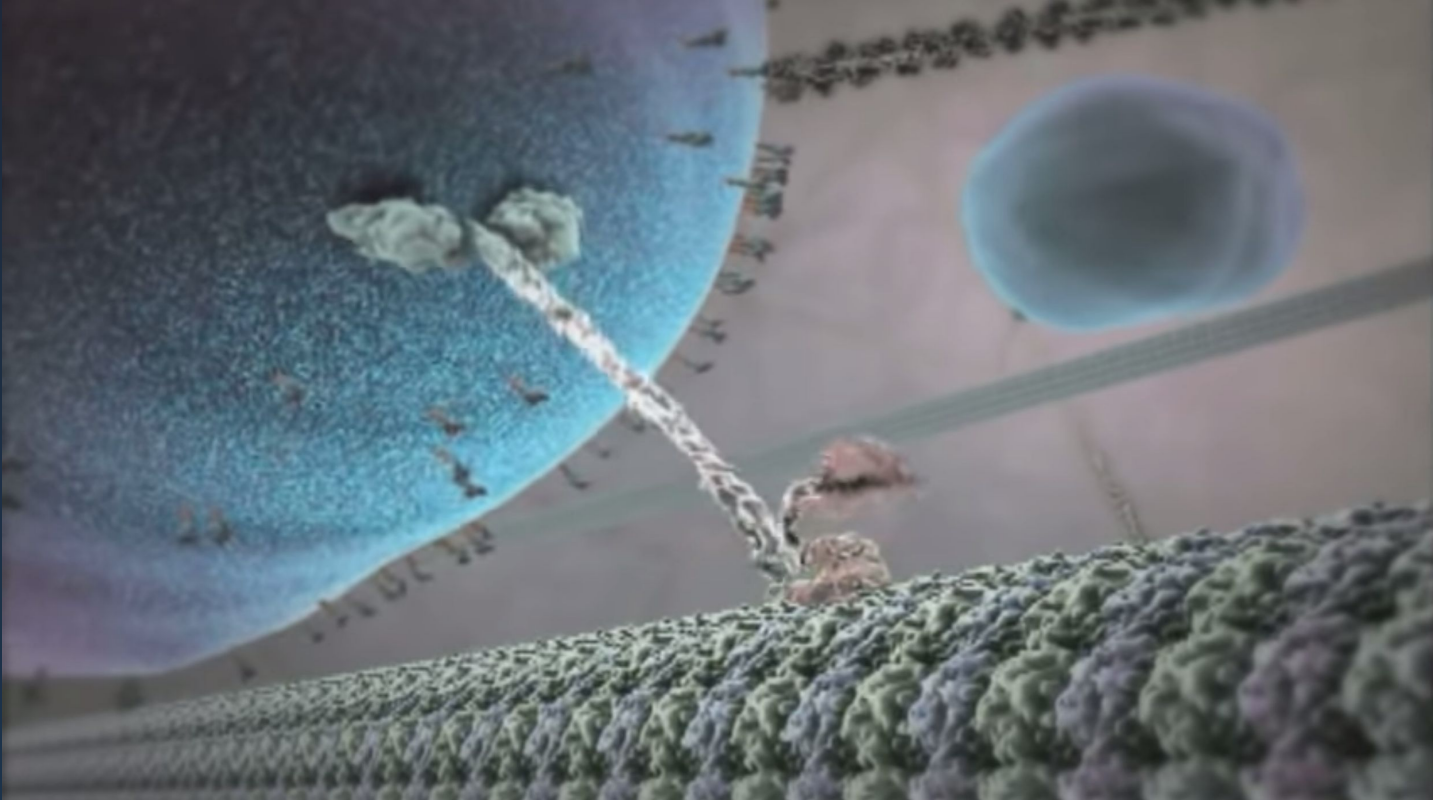
WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 3: Data, Information, Knowledge



Molecular Motors are “Systems”



<https://www.youtube.com/watch?v=wJyUtbn0O5Y>

DNA is a System



DNA

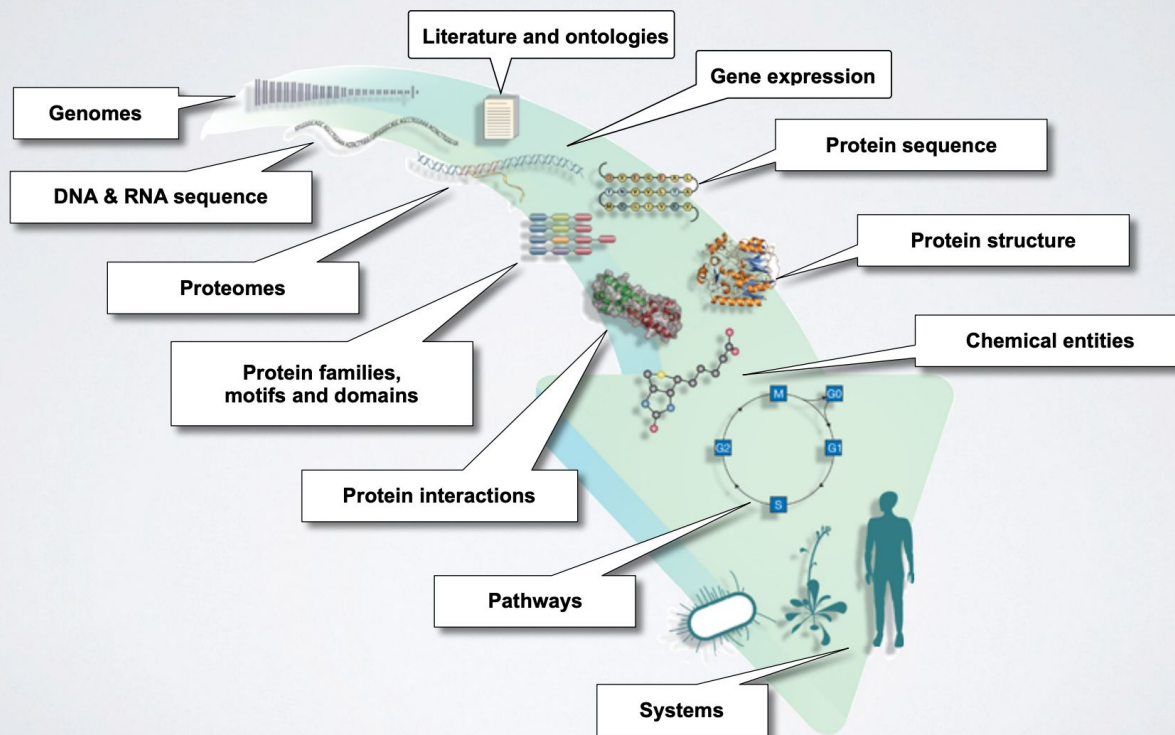


deoxyribonucleic acid

Rise of Complexity: “Omes” are Systems

Genome	The full complement of genetic information both coding and noncoding in an organism
Proteome	The complete set of proteins expressed by the genome in an organism
Transcriptome	The population of mRNA transcripts in the cell, weighted by their expression levels as transcripts copy number
Metabolome	The quantitative complement of all the small molecules present in a cell in a specific physiological state
Interactome	Product of interactions between all macromolecules in a cell
Phenome	Qualitative identification of the form and function derived from genes, but lacking a quantitative, integrative definition
Glycome	The population of carbohydrate molecules in the cell
Translatome	The population of mRNA transcripts in the cell, weighted by their expression levels as protein products
Regulome	Genome wide regulatory network of the cell
Operome	The characterization of proteins with unknown biological function
Synthetome	The population of the synthetic gene products
Hypothome	Interactome of hypothetical proteins

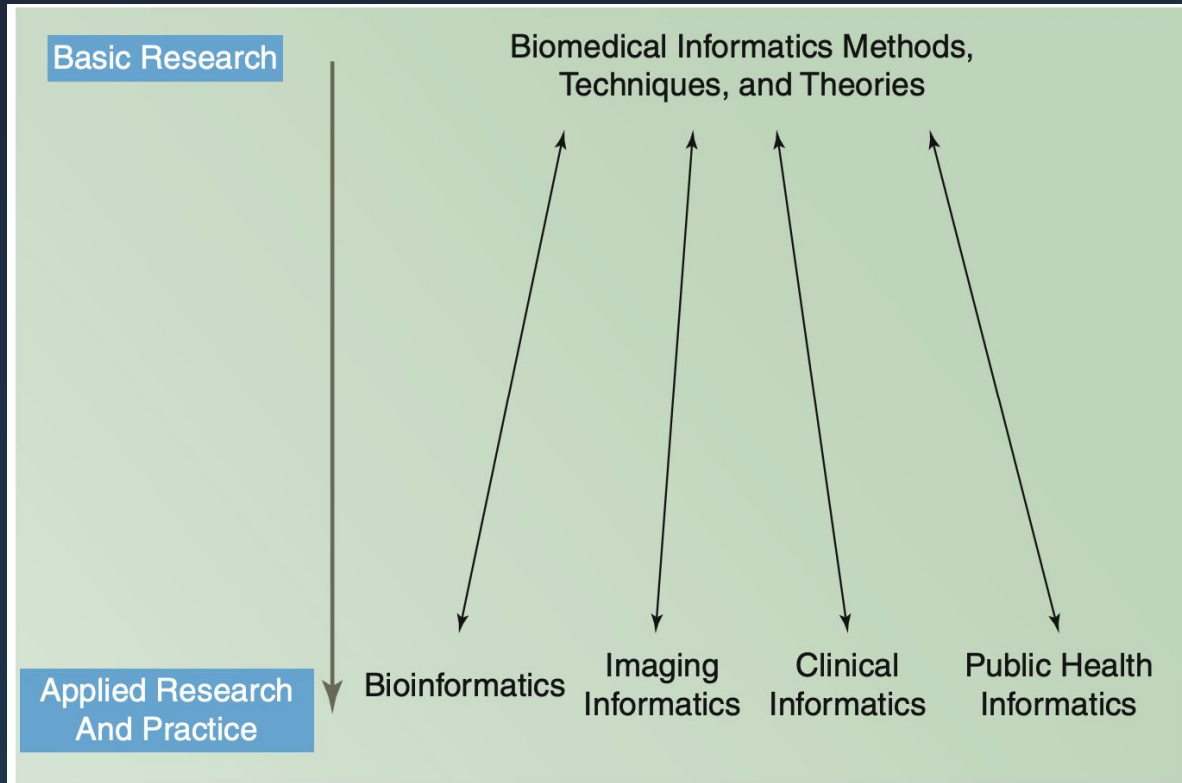
Biomedical Informatics is the Study of These “Systems”



How is Biomedical Informatics Different from CS

- **1970s: Term medical computer science used for applying CS methods to medicine**
- **Medicine proved to be a rich testbed for CS research**
- **Led to new computing insights, methods, and models**
- **Shift in thinking:**
 - **Not just “using computers”**
 - **But understanding information, reasoning, and decision-making in medicine**

How is BMI Different from Bioinformatics



Information Science and Information Theory

Information Science

- **Originated in library science**
- **Focus: managing paper-based and electronic information**
- **Many ideas later absorbed by cognitive science**

Information Theory

- **Mathematical theory of communication**
- **Powerful for signals and networks**
- **Neutral, tool-oriented terms**
- **Often associated with bioengineering applications**
- **Computers viewed as instruments, not the research focus**

Informatics

- **Informatics is the science of information, not simply the science of computers.**
- **Informatics is a young discipline — it has existed for only a little over 50 years.**
- **The term “informatics” comes from information + automatics (Steinbuch, 1957).**
- **It was coined to describe the science of automatic information processing.**
- **Informatics researches in and with computational methods, so it uses computers extensively.**
- **Informatics researchers don't necessarily build computers or algorithms, although it is advisable.**

Biomedical Informatics

Medical Informatics (1970s–1980s)

- **Emphasized information over machines - Broader than medical computing (statistics, records, knowledge)**
- **Concerns: “Medical” too physician-centric**

Health Informatics

- **Broader clinical and public health focus**
- **Risked excluding biomedical research**

Biomedical Informatics (BMI)

- **Emerged in late 1990s–2000s**
- **Unified clinical care + biology + research**
- **Became the dominant term globally**

Biomedical Informatics

- **Biomedical informatics is an interdisciplinary field that focuses on the effective use of biomedical data, information, and knowledge.**

It supports key aims such as:

- **scientific inquiry**
- **problem solving**
- **decision making**
- **The ultimate motivation is to improve human health (Shortliffe, 2011).**
- **Central idea: use machines to support and amplify human intelligence (Holzinger, 2013).**

Biomedical Informatics

- **Encompasses health, clinical practice, and biomedical research**
- **Integrates:**
 - **Computer science**
 - **Statistics**
 - **Decision sciences**
 - **Cognitive & information sciences**
 - **Social & management sciences**

Central idea:

- **BMI is not about computers it is about data, information, knowledge, and decisions in biomedicine**

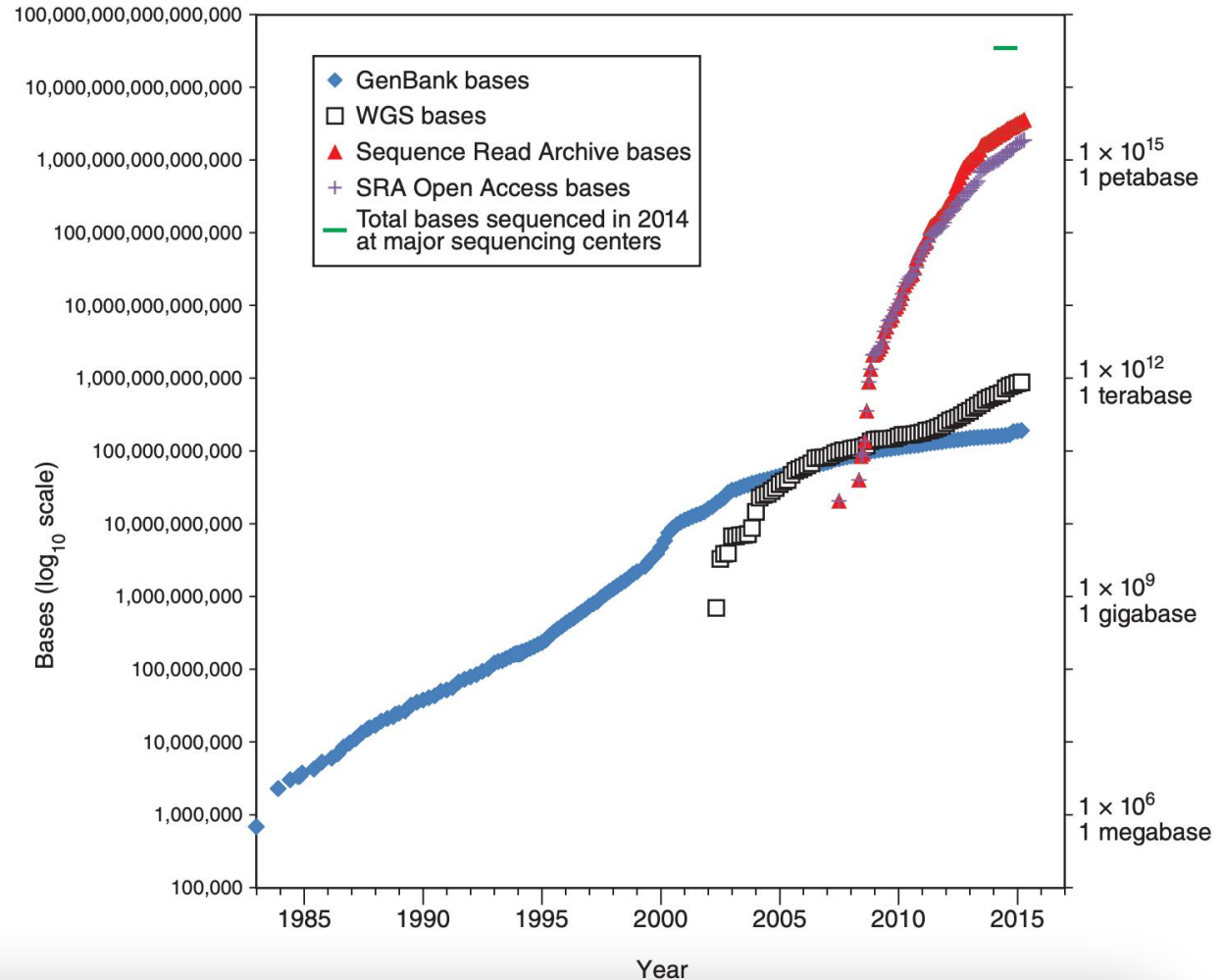
What is Biomedical Data?

- **Biomedical data describe observations made about a patient or biological system**
- **Each data point is context-dependent, not just a number**
- **Meaning arises from who, what, when, and how the observation was made**
- **The subject (patient): The individual to whom the data refers**
- **The parameter observed: What is being measured or assessed**
- **Examples: liver size, urine glucose, heart size on X-ray, past medical history**

What is Biomedical Data?

- **The value of the parameter: The actual observed value**
- **Examples: weight = 70 kg, temperature = 98.6°F, profession = steel worker**
- **The time of observation: When the measurement was made**
- **Example: 2:30 AM on 14-Feb-2013**
- **The method of observation: How the data was obtained**
- **Examples: patient report, thermometer, urine dipstick, lab instrument**

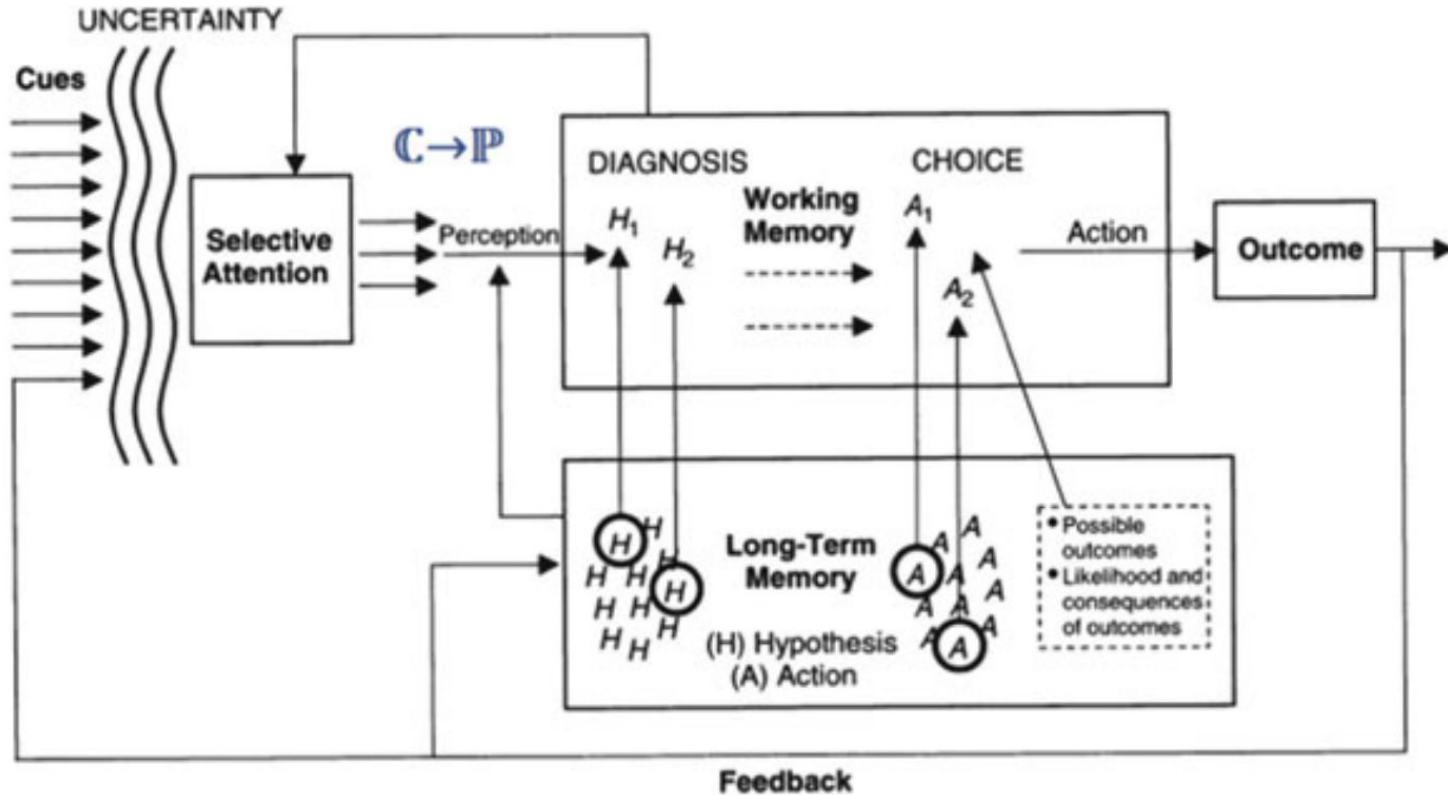
Growth of Biomedical Data: Eroom's Law



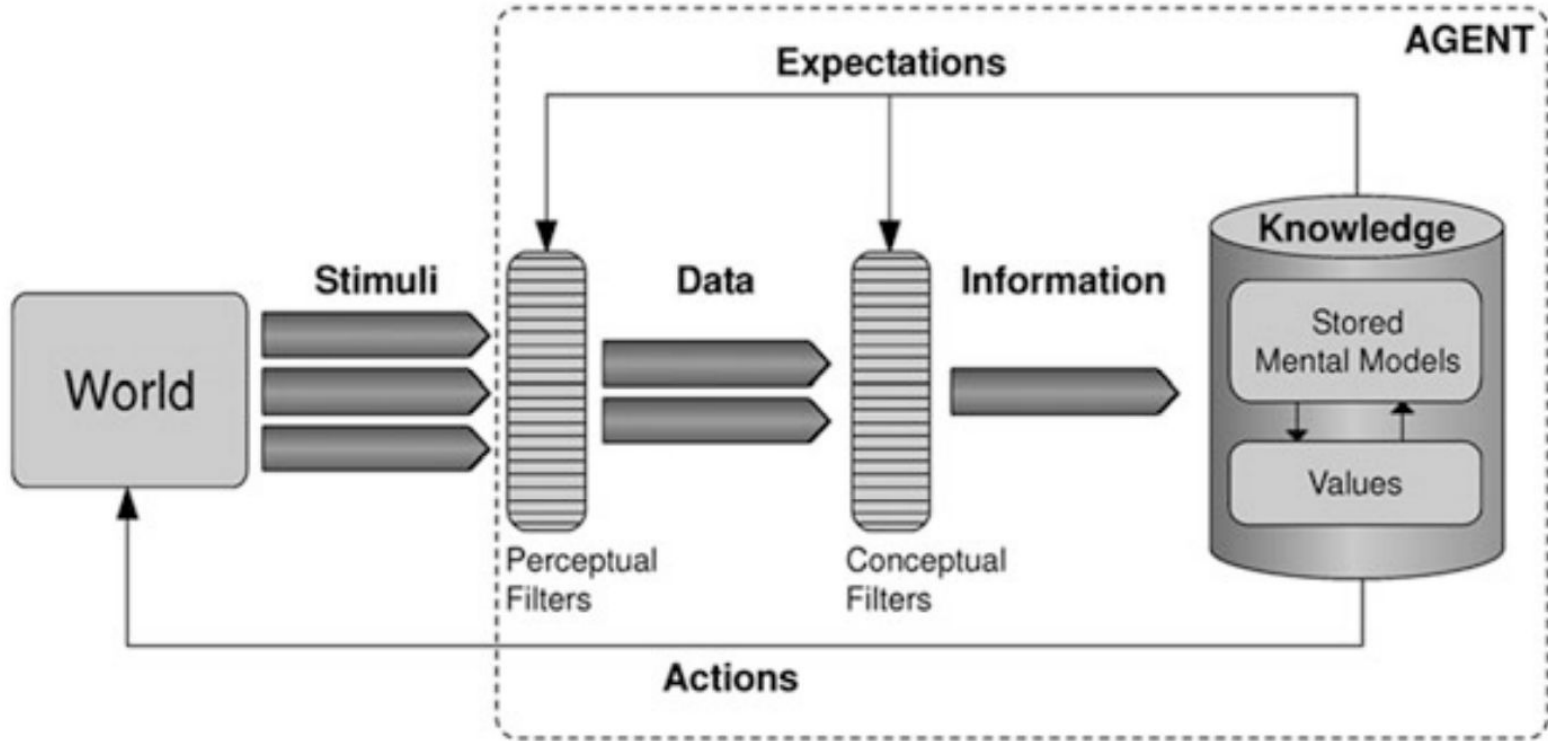
What is Information? Human Information Processing Loop

- **Real-world physical data are sensed by humans**
- **Data are transformed into information via perceptual filters**
- **Selective attention determines what information is processed**
- **Information forms hypotheses in working memory**
- **Hypotheses are shaped by existing mental models**
- **These models are stored in long-term memory**
- **Expectations depend on prior experience and knowledge**
- **The best hypothesis (subjectively) is selected**
- **Selected hypothesis guides action or decision**
- **Actions generate new data**
- **New data are perceived as new information**
- **The process repeats continuously (feedback loop)**

A Cognitive Model of Decision Making in Clinical Scenarios



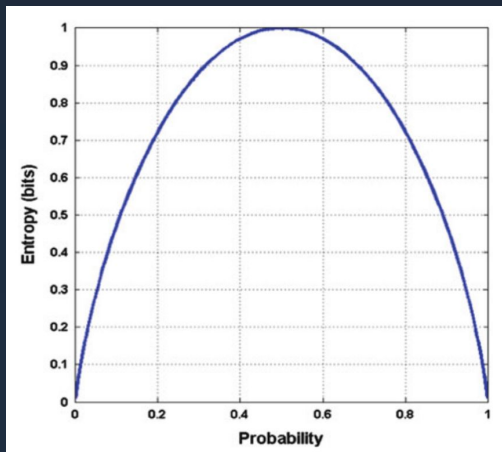
Human Information Processing Model



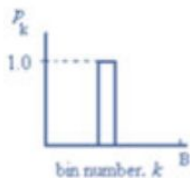
Knowledge = Expectation + Uncertainty (Entropy)

- Knowledge can be viewed as a set of expectations used to interpret data
- Uncertainty is unavoidable in biology/medicine; we need measures of uncertainty
- Example: Entropy as a quantifier of uncertainty/information

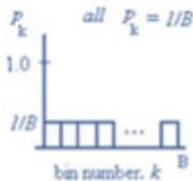
Shannon's Entropy:



Entropy Can Be Used to Quantify Structure



$$H_B = - \sum_{k=1} p_k \log_2 p_k = -1 * \log_2(1) = 0$$

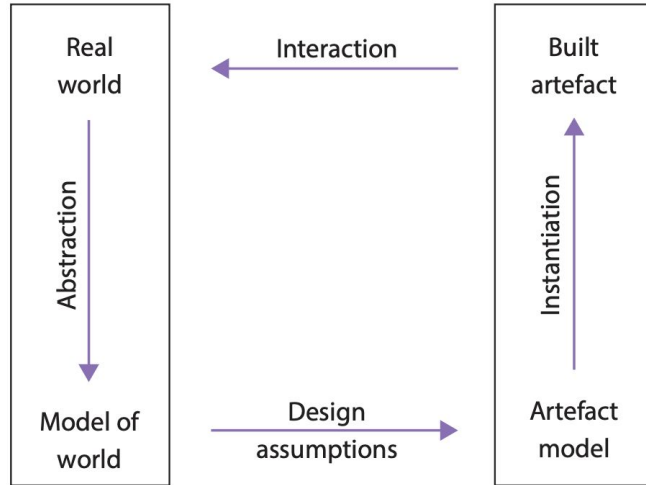


$$H_B = - \sum_{k=1}^B \frac{1}{B} \log_2 \frac{1}{B} = \log_2(B)$$



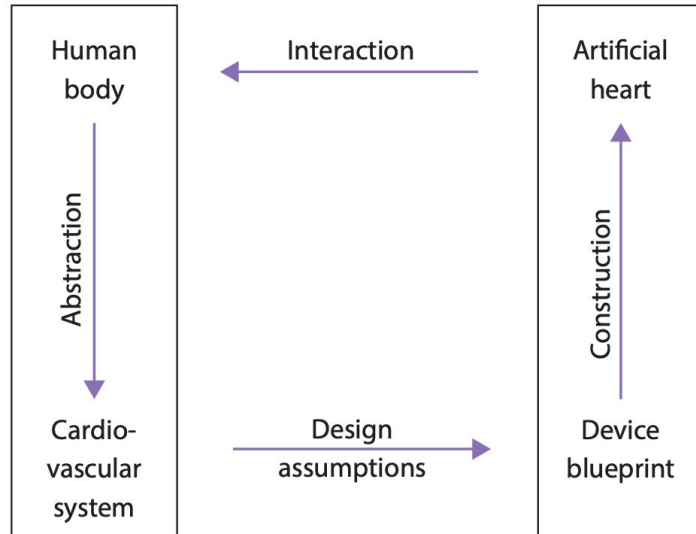
$$H = H_{\min} = 0 \quad H = H_{\max} = \log_2 N$$

What are Models?

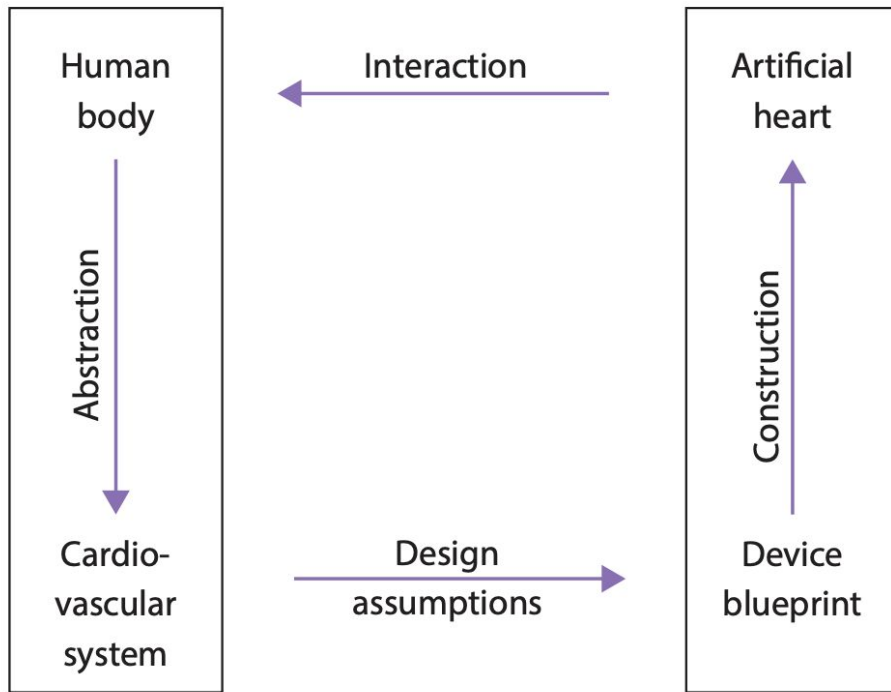


**Models Reduce
Complexity of Systems**

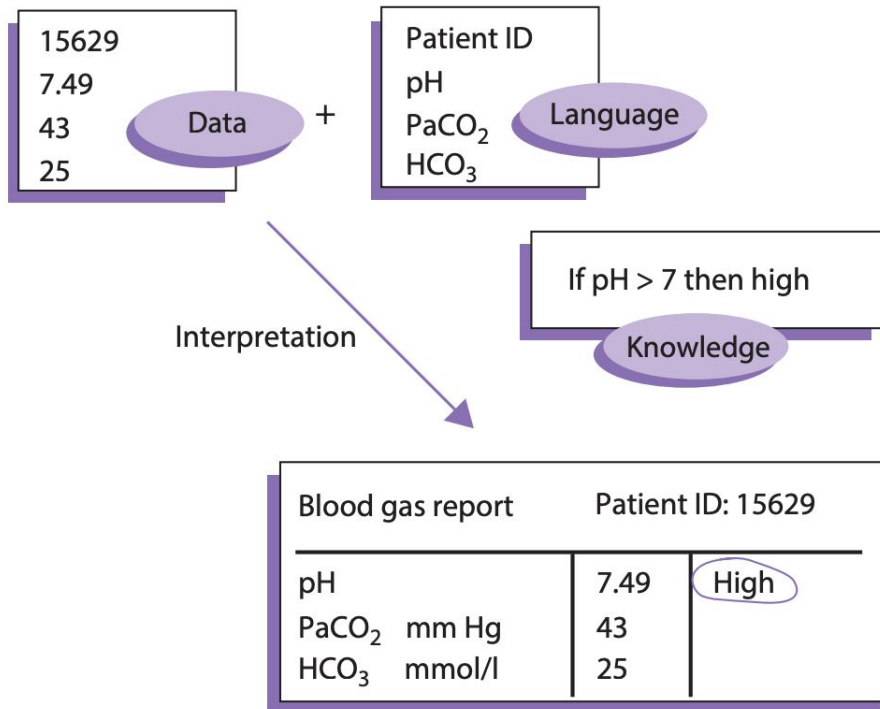
Act as Templates to be Instantiated to Simulate Systems



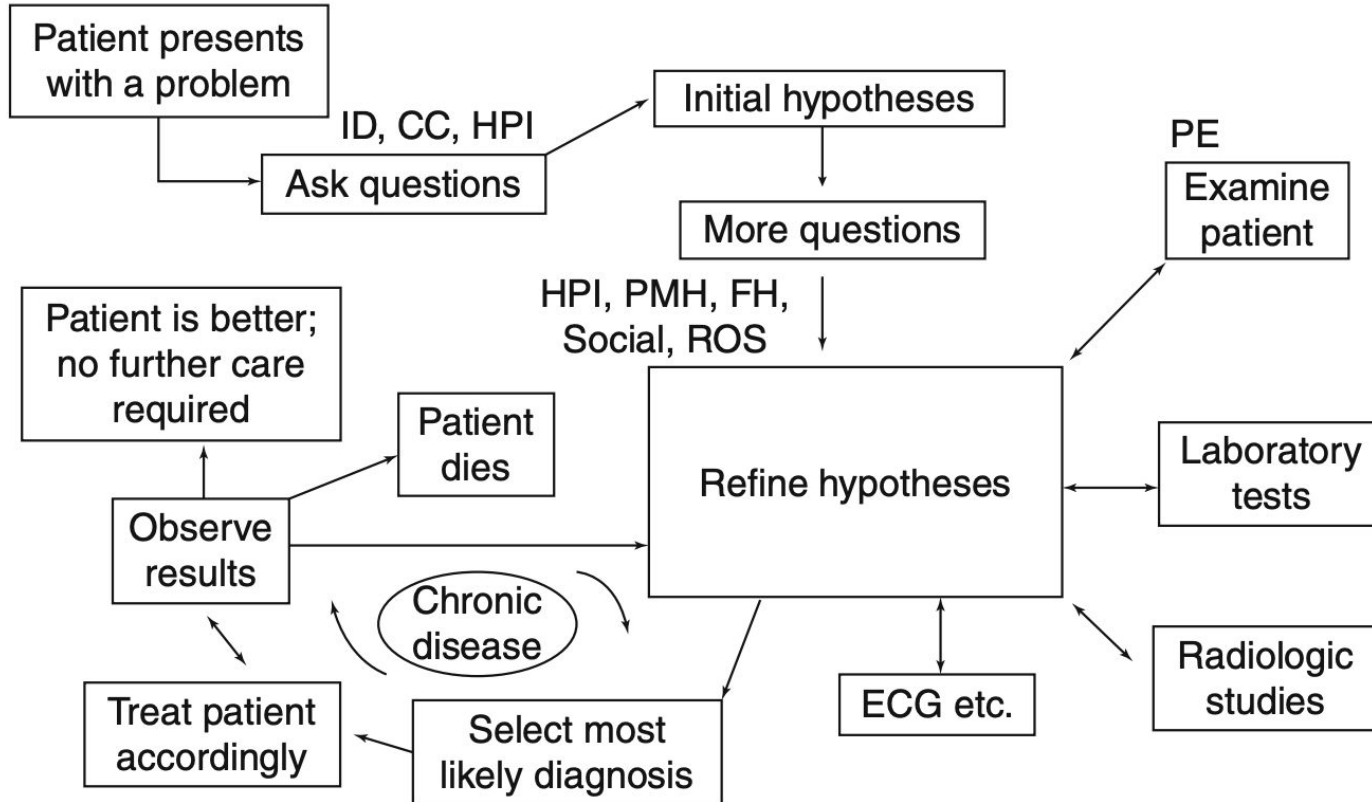
Models Create Real World Systems



Link Between Language, Data, Knowledge, Model



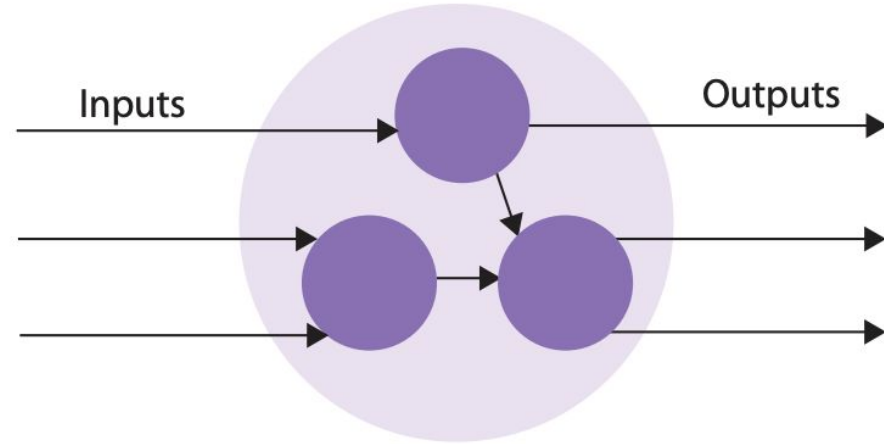
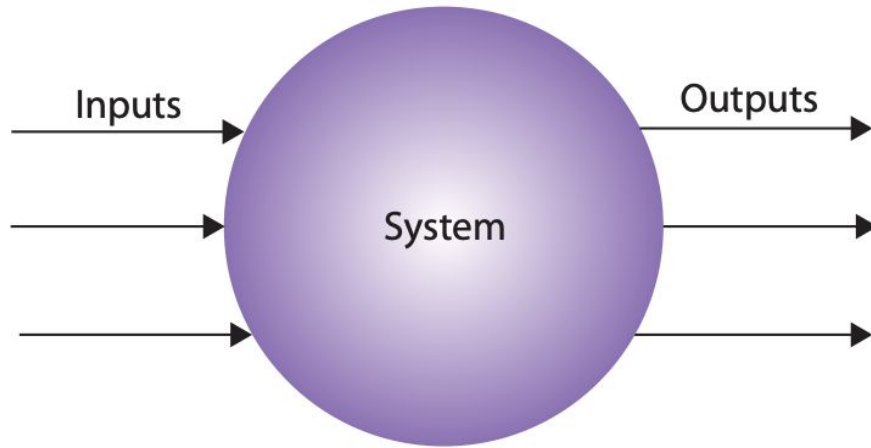
Link Between Language, Data, Knowledge, Model



What is a Learning Health System Model?

- Routine care generates data → evidence → improved care (continuous feedback loop)
- Alignment needed
- science + informatics
- incentives
- culture
- governance/ethics
- Goal: best care at lower cost, continuously improving

Learning Health System



Learnings

- Expect **adaptation + feedback loops**
- Monitor early and continuously (surveillance + rapid response)
- Use multi-pronged strategies (combination therapy, stewardship, infection control)
- Update models with real-world data (continuous learning)
- Build policies assuming **unknown unknowns** (resilience, not certainty)

Bottom line: *Act, but design interventions as experiments in a complex system.*

How Systems Work: Example from Antibiotic Resistance

What happened

- Antibiotics created huge optimism: infectious disease looked “solvable.” Over time, bacteria evolved resistance (e.g., penicillinase producers).
- Antibiotics **selected for resistant survivors**, who then dominated populations.
- Some resistant infections now disrupt entire hospitals.

Why it happens (systems lens)

- Every intervention is based on a **simplified model** of reality. We act assuming “everything else is equal” → but systems never behave that cleanly. We rely on the **closed-world assumption**: “we know enough to act.”
- Resistance is not just a biology problem it’s a **predictable systems outcome** of intervention under **incomplete knowledge**.